

DEEP REINFORCEMENT LEARNING

365.382 (SS26)

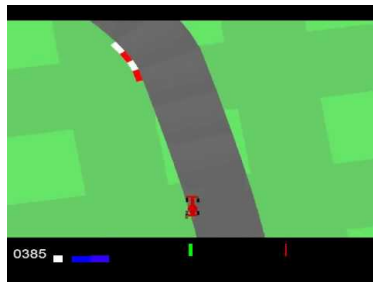
Assignment III: Learning to Drive with Imitation Learning



LIT AI LAB
Institute for Machine Learning

Exercise

- Objective: Given a dataset with expert trajectories, learn a policy that imitates the expert.
- Tasks:
 1. Implement neural network architecture
 2. Train an initial policy with behavioral cloning
 3. Implement the DAgger algorithm
 4. Train a policy with DAgger



Task 1: Neural Network Architecture

- Network input is fixed (you get the correct input from the provided dataset)
- You are free in the design of your network (but be careful with file size limit of 20MB for uploaded models)
 - ☐ Network architecture (type, number of layers and units, ...)
 - ☐ Activation functions
 - ☐ ...

Task 2: Behavioral Cloning

- Implement a simple training and validation loop
- Training and validation sets are provided and consist of episodes generated by an expert agent
- Choose an appropriate loss function
- Decide on an optimizer and tune its hyper-parameters
- Optional:
 - ☐ Learning rate schedule
 - ☐ Choose your own logging framework (e.g. Weights & Biases)
 - ☐ Store model weights and other parameters to continue learning

Task 3: Implement DAgger

- You can initialize $\hat{\pi}_1$ randomly or with a previously trained BC policy
- Try different values for β
- You can anneal β during training, i.e.
 - ☐ Start with high beta to explore with expert
 - ☐ Reduce β during training to let the training policy explore
 - ☐ or the other way around

Submission

- Submission server: <https://apps.ml.jku.at/challenge>
- Export your trained agent as ONNX file
- Upload ONNX file to submission server
 - ☐ Code submission closes on **Friday, May 22nd, 18:00**.
 - ☐ Max model file size: **50MB**.
- Submit your **code**, **best model** and a short **report** to Moodle
 - ☐ Report the performance on your local setting.
 - ☐ Moodle submission stays open 24h after the challenge closes.
- Please upload a zip file named `k<studentid>.zip` containing:
 - ☐ `code.ipynb` / `code.py`
 - ☐ `model.onnx`
 - ☐ `report.pdf`

Evaluation

■ < 100 : 0 pts

■ ≥ 100 : 2 pts

■ ≥ 400 : 4 pts

■ ≥ 600 : 6 pts

■ ≥ 700 : 8 pts

Presentations

- If you are willing to give a short presentation of your work, fill out the poll Model that closes Sunday evening after the submission deadline.
- Students selected to present will receive +2 bonus points.